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Okamura et al.

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(54) **DATA CENTER SYSTEM, INTER-BASE
WORKLOAD CONTROL METHOD, AND
INTER-BASE WORKLOAD CONTROL
SYSTEM**

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H04L 67/1012 (2022.01)

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CPC **H04L 67/1012** (2013.01); **H04L 67/1008**
(2013.01)

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USPC 709/203
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(56) **References Cited**

U.S. PATENT DOCUMENTS

8,261,275 B2 * 9/2012 Johnson G06F 9/4881
719/318
8,312,229 B2 * 11/2012 Bloks G06F 9/4887
710/5
8,335,859 B2 * 12/2012 Nagpal G06F 9/5094
709/239
8,732,267 B2 * 5/2014 Banerjee G06F 9/5072
709/217
9,886,316 B2 * 2/2018 Belady G06F 9/5094
(Continued)

FOREIGN PATENT DOCUMENTS

JP 2021-189845 A 12/2021

OTHER PUBLICATIONS

Extended European Search Report received in corresponding Euro-
pean Application No. 24161463.5 dated Aug. 21, 2024.

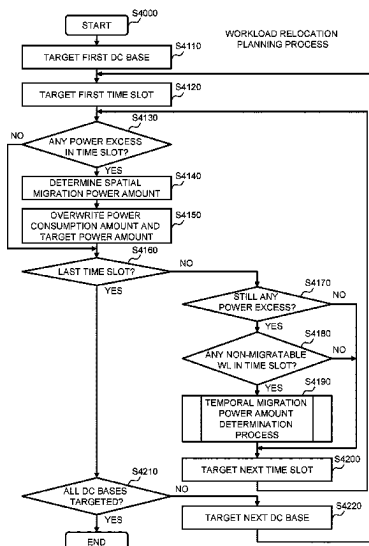
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(57) **ABSTRACT**

Power demand at each base is adjusted so as to improve a
renewable energy utilization ratio at all bases. An inter-base
workload control system manages an amount of excess
power obtained by subtracting a power supply amount of the
renewable energy power supply from a power consumption
amount associated with execution of a workload in a future
time range at the bases, spatial migratable time range
information on a spatial migratable time range where spatial
migration of migrating the workload in a future time range
at the bases to another base is possible and temporal migrat-
able time range information on temporal migration of delay-
ing execution of the workload in the future time range at the
bases within the same base and migrating the workload to
another time range and a predicted amount of power con-
sumption through execution of the workload in the future
time range at the bases.

9 Claims, 13 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

10,048,979	B2 *	8/2018	Vincent	G06F 9/4856
10,067,547	B2 *	9/2018	Castro-Leon	G06F 1/266
10,185,586	B2 *	1/2019	Kumeta	G06F 9/5094
10,776,147	B2 *	9/2020	Ovesea	G06F 9/4856
10,852,805	B2 *	12/2020	Magcale	G06F 11/3006
11,656,911	B2 *	5/2023	Bahramshahry	G06F 9/505
				718/103
11,823,294	B1 *	11/2023	Kaneko	G06Q 10/04
12,099,873	B2 *	9/2024	Grimshaw	G06F 9/4887
2004/0024861	A1 *	2/2004	Coughlin	H04L 67/1008
				709/224
2005/0034128	A1 *	2/2005	Nagashima	G06F 9/4881
				718/100
2014/0365402	A1	12/2014	Belady et al.	
2016/0378532	A1	12/2016	Vincent et al.	
2019/0235895	A1	8/2019	Ovesea et al.	
2024/0104478	A1 *	3/2024	Ozawa	G06Q 10/06375
2024/0155026	A1 *	5/2024	Sreenivasan	H04L 67/1012
2024/0411594	A1 *	12/2024	Grimshaw	G06F 9/4887

* cited by examiner

FIG. 1

DATA CENTER SYSTEM

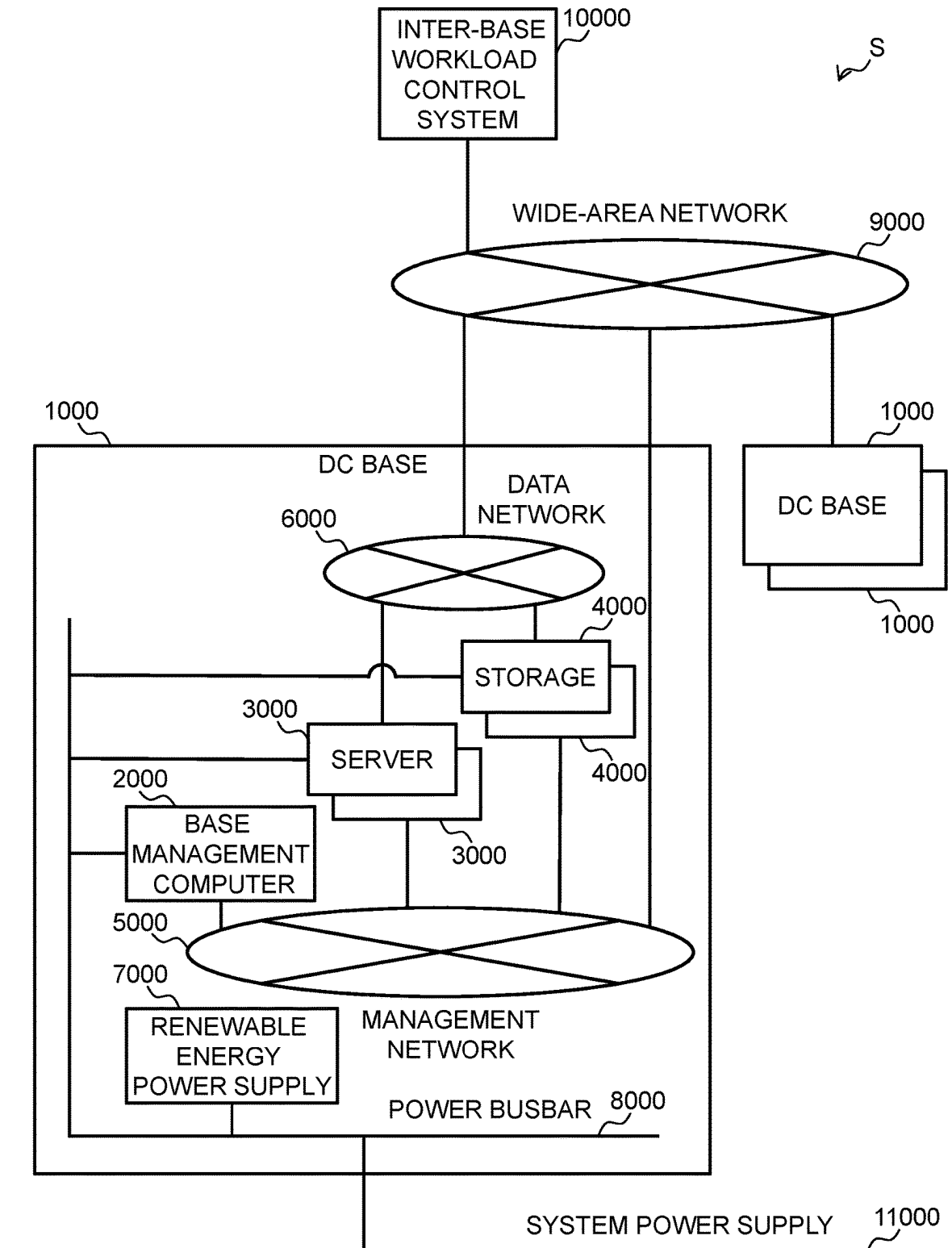


FIG. 2

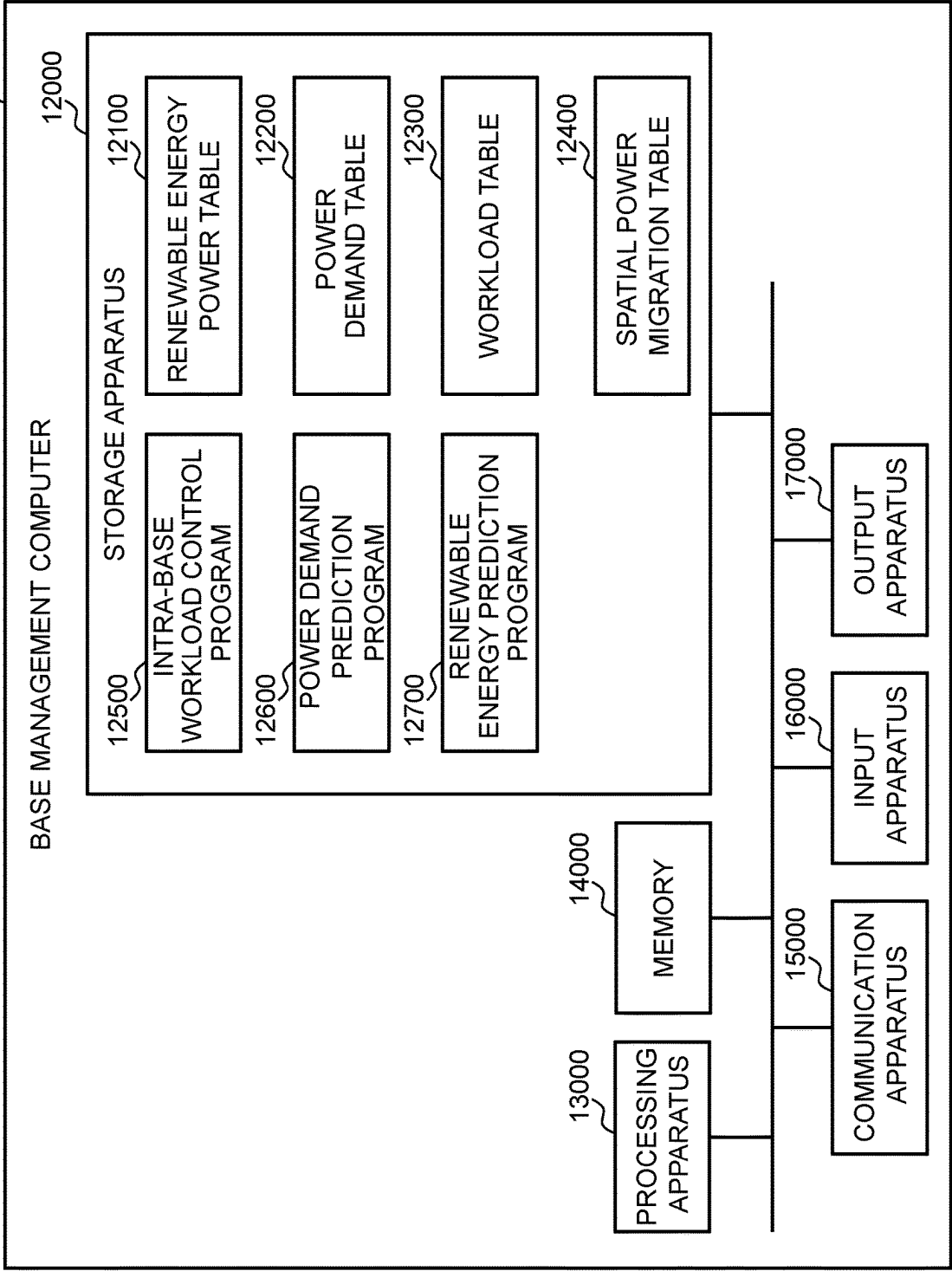


FIG. 3

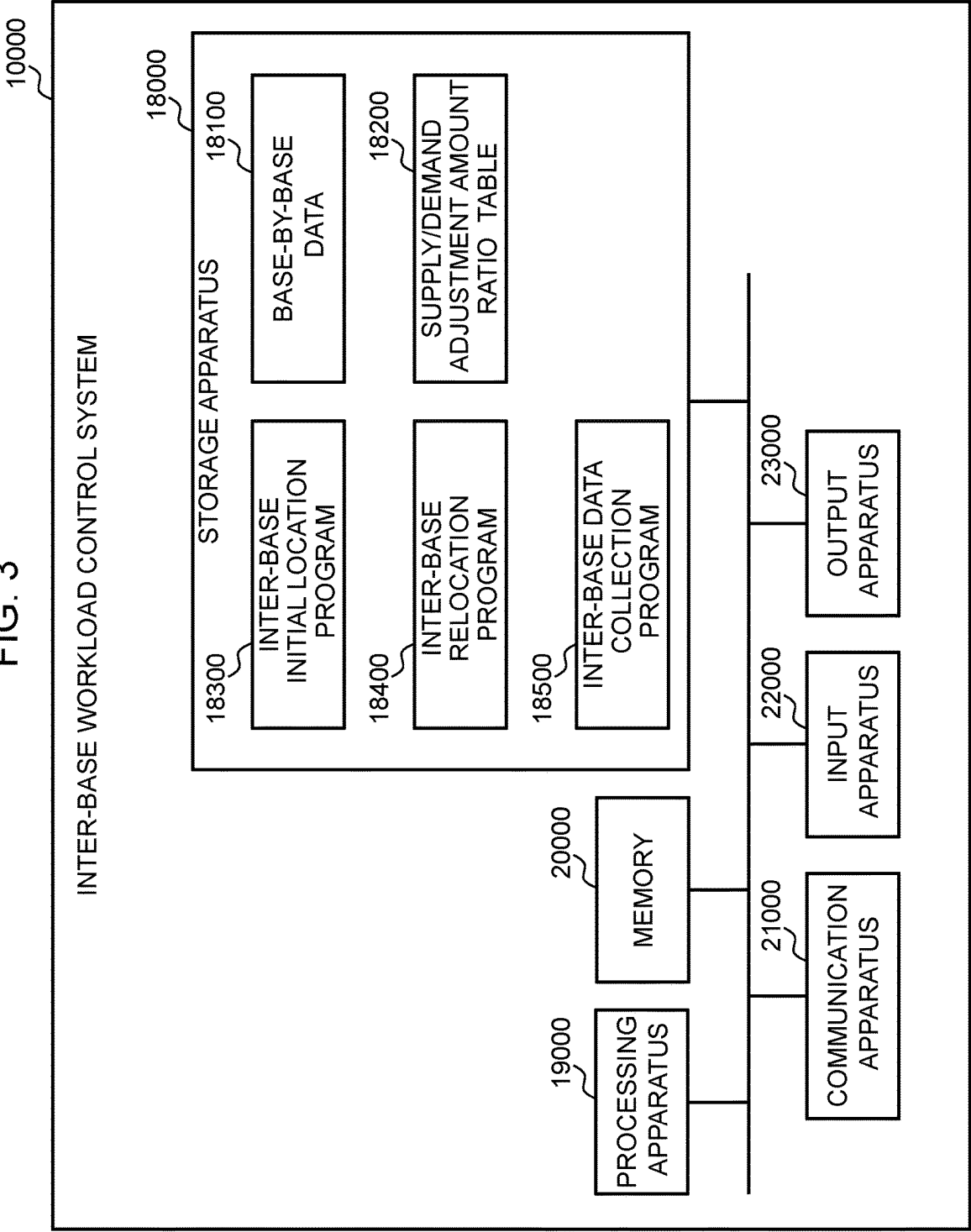


FIG. 4

12100

RENEWABLE ENERGY POWER TABLE

12110 TIME SLOT ID	12120 CLOCK TIME	12130 RENEWABLE ENERGY SUPPLY AMOUNT PREDICTION	12140 RENEWABLE ENERGY SUPPLY AMOUNT ACTUAL MEASUREMENT
01	2018/4/1 10:00	3.0	2.5
02	2018/4/1 11:00	5.3	4.0
03	2018/4/1 12:00	10.2	12.2
04	2018/4/1 13:00	9.1	n/a
...	...	...	...

FIG. 5

12200

POWER DEMAND TABLE

12210	12220	12230	12240	12250
TIME SLOT ID	CLOCK TIME	DC POWER CONSUMPTION AMOUNT PREDICTION	DC POWER CONSUMPTION AMOUNT ACTUAL MEASUREMENT	POWER CONSUMPTION TARGET VALUE
01	2018/4/1 10:00	10.1	10.9	8.5
02	2018/4/1 11:00	7.8	6.5	9.0
03	2018/4/1 12:00	6.3	7.0	4.7
04	2018/4/1 13:00	9.6	n/a	n/a
...	...	...	...	

FIG. 6

WORKLOAD TABLE

12305 WORKLOAD ID	12310 POWER CONSUMPTION AMOUNT PREDICTION	12315 POWER CONSUMPTION AMOUNT ACTUAL MEASUREMENT	12320 INPUT TIME	12325 EXECUTION SCHEDULE	12330 CHANGED EXECUTION SCHEDULE
01	4.7	n/a	2018/4/1 10:05	n/a	n/a
02	8.2	n/a	2018/4/1 11:30	n/a	n/a
03	0.2	0.2	2018/4/1 13:40	2018/4/1 14:00	n/a
04	1.0	n/a	2018/4/1 14:01	2018/4/1 15:05	2018/4/1 16:01
...	...	...	...	...	...

12335 DELAYABLE TIME	12340 EXECUTABLE BASE	12345 SPATIAL MIGRATABLE TIME RANGE	12350 SPATIAL NON-MIGRATABLE TIME RANGE	12355 EXECUTION STATUS
n/a	1,2	0:00-7:59, 17:01-23:59	8:00-17:00	BEFORE EXECUTION
n/a	1,2,3,4,5	0:00-23:59	n/a	EXECUTION IN PROGRESS
0mins	1	n/a	n/a	EXECUTED
480mins	1,4,5	n/a	n/a	EXECUTION IN PROGRESS
...	...	...	...	...

FIG. 7

SPATIAL POWER MIGRATION TABLE

12400
↙

12410 TIME SLOT ID	12420 CLOCK TIME	12430 DC BASE 2	...	DC BASE N
01	2018/4/1 11:00	4.0		5.0
02	2018/4/1 12:00	0.0		0.0
03	2018/4/1 13:00	1.0		0.1
04	2018/4/1 14:00	2.1		0.0
...	...	...		

FIG. 8

SUPPLY/DEMAND ADJUSTMENT AMOUNT RATIO TABLE

18200
↙

18210 TIME SLOT ID	18220 CLOCK TIME	18230 DC BASE 1	...	DC BASE N
01	2018/4/1 11:00	0.1		0.2
02	2018/4/1 12:00	0.2		0.5
03	2018/4/1 13:00	0.5		0.1
04	2018/4/1 14:00	0.0		0.7
...	...	...		

FIG. 9

WORKLOAD CONTROL PLANNING PROCESS

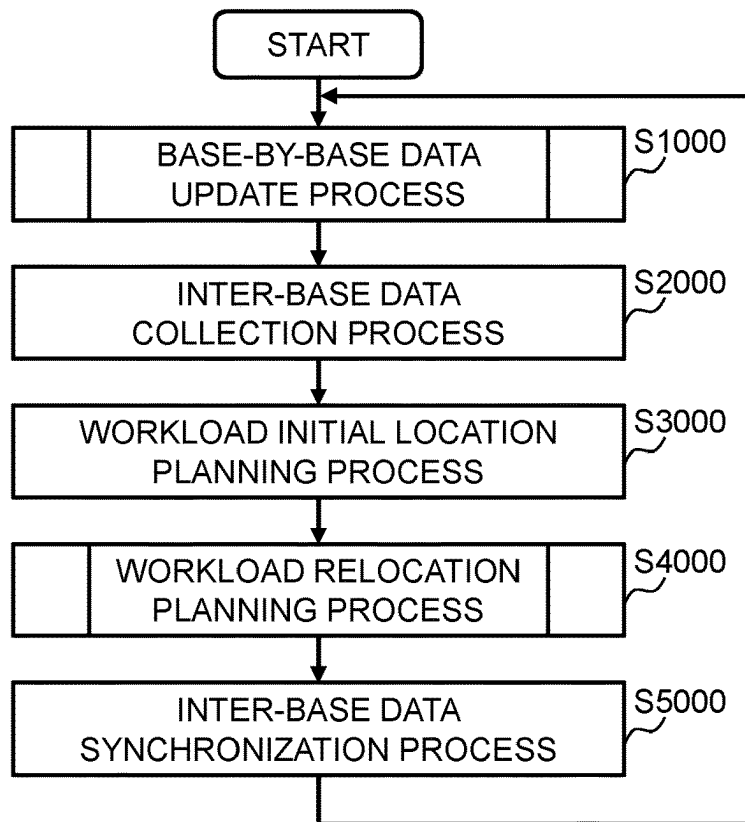


FIG. 10

BASE-BY-BASE DATA UPDATE PROCESS

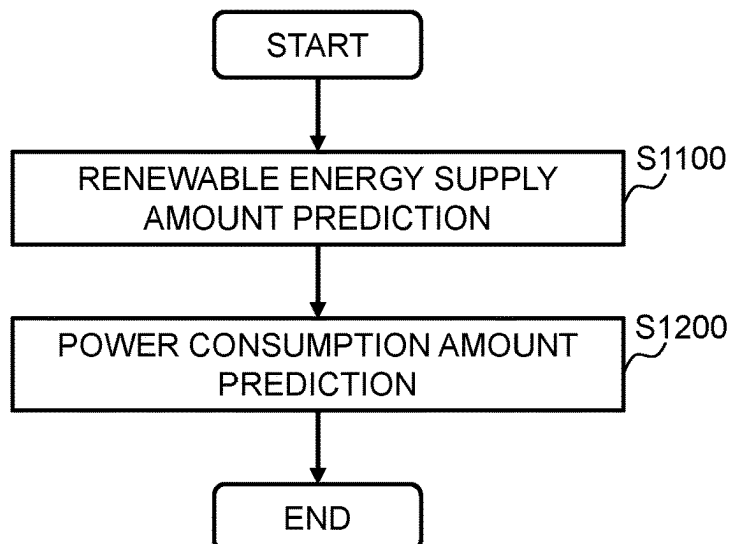
S1000
↙

FIG. 11

WORKLOAD CONTROL EXECUTION PROCESS

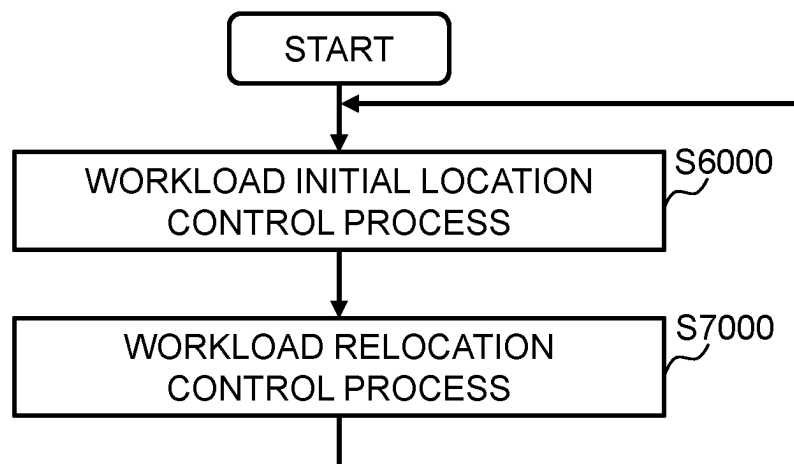


FIG. 12

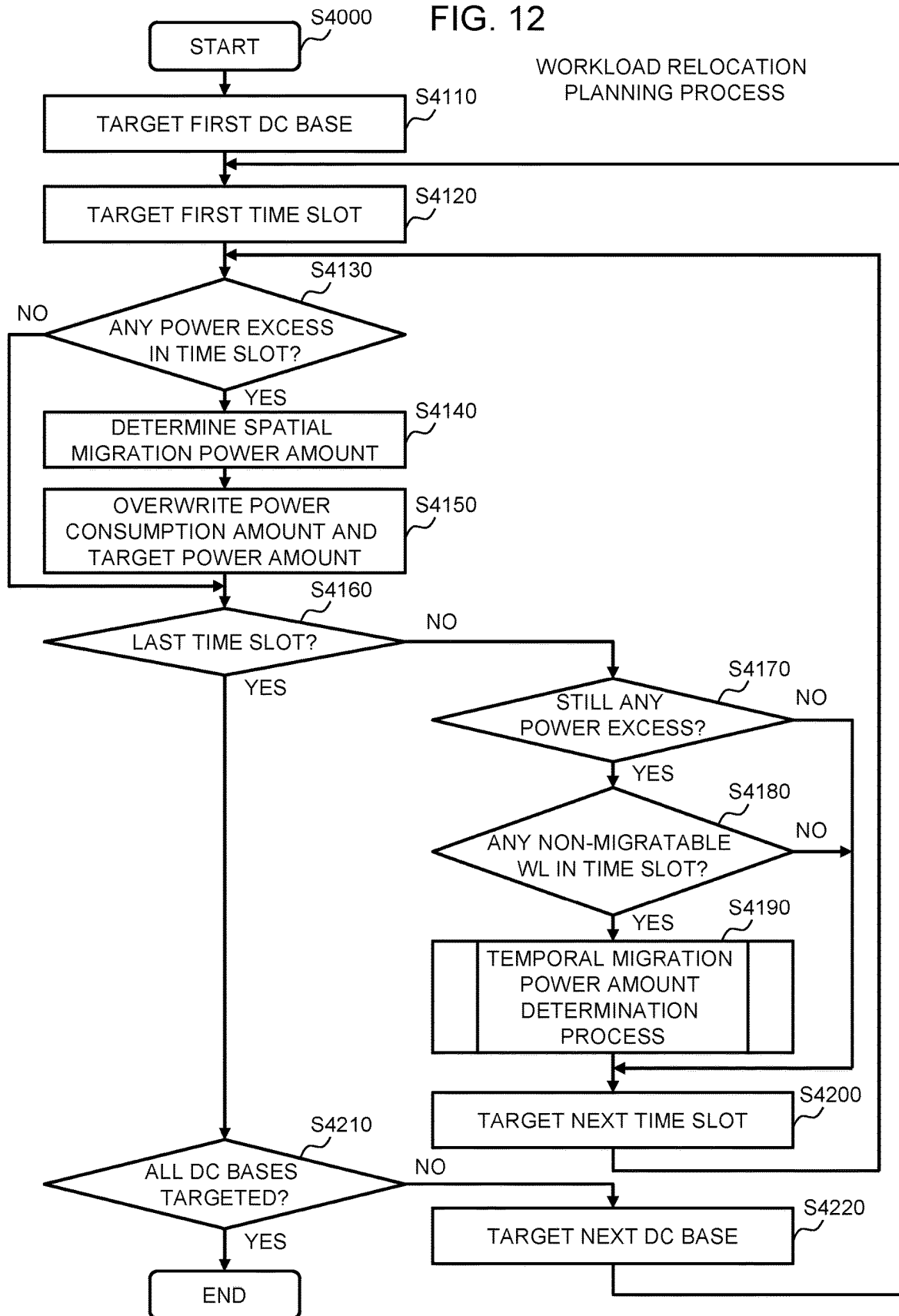


FIG. 13

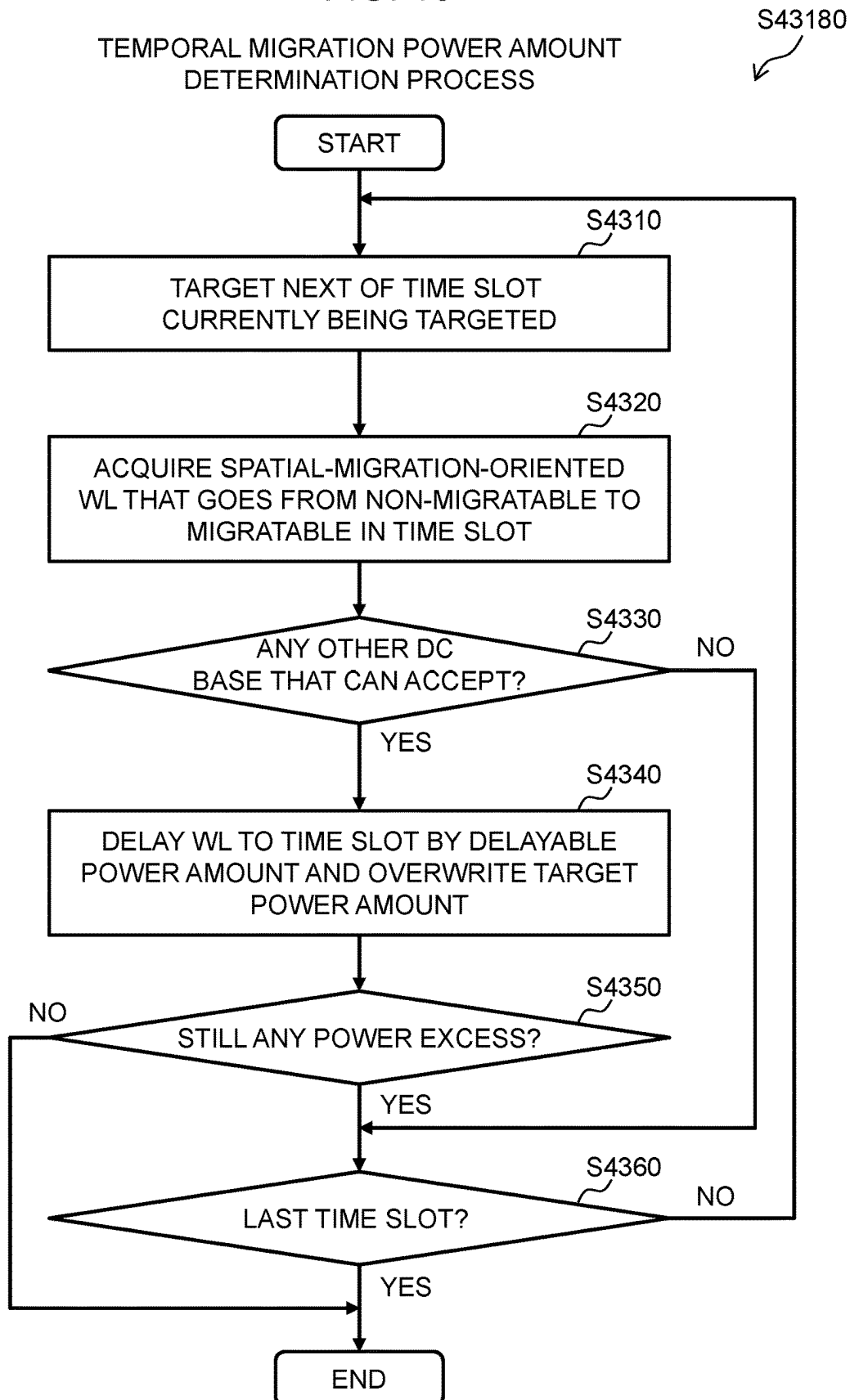
TEMPORAL MIGRATION POWER AMOUNT
DETERMINATION PROCESS

FIG. 14

24000

WORKLOAD INPUT SCREEN

■ LOCATABLE DC BASE	1,2,3	24100
■ DELAYABLE TIME RANGE	n/a	24200
■ SPATIAL MIGRATABLE TIME RANGE	6:00-21:00	24300
■ SPATIAL NON-MIGRATABLE TIME RANGE	0:00-5:59, 21:01-23:59	24400
■ SOURCE CODE	/home/workload	24500
	ENTER	24600

DATA CENTER SYSTEM, INTER-BASE WORKLOAD CONTROL METHOD, AND INTER-BASE WORKLOAD CONTROL SYSTEM

CROSS-REFERENCE TO RELATED APPLICATION

This application is based upon and claims the benefit of priority of the prior Japanese Patent Application No. 2023-86288, filed on May 25, 2023, the entire contents of which are incorporated herein by reference.

BACKGROUND

The present invention relates to a data center system, a DC inter-base workload control method and a DC inter-base workload control system.

DESCRIPTION OF THE RELATED ART

Attention has been attracted to so-called decarbonization, which aims to move away from fossil fuels so as to prevent emissions of greenhouse gases such as carbon dioxide, which causes global warming. In this respect, many information processing apparatuses and communication equipment to execute predetermined processing loads (workloads) are set in data centers (DCs), and operating these apparatuses and equipment requires a large amount of electric power. Therefore, attempts are being made to achieve decarbonization by providing such electric power from renewable energy sources.

In this case, while it is important to keep a ratio of renewable energy to power consumption (renewable energy utilization ratio), it is preferable to keep this renewable energy utilization ratio in order to achieve cleaner DCs with finer time granularity (e.g., hourly rather than daily).

Another problem is that supplies of renewable energy depend on factors such as weather, and so energy supplies are quite unstable, and as the dependency on renewable energy increases, power providers need to be provided with more coordinating power (power necessary to match demand and supply of electricity) so that the power providers perform stable power supplies, and this constitutes a burden on the power providers.

Therefore, there are attempts to control power consumed at DCs by temporally controlling workloads executed at DCs or by spatially controlling the workloads between a plurality of DC bases (collection of DCs within the same area) to control power consumed at DCs and thereby improve the renewable energy utilization ratio or utilize the DCs as coordinating power for the power system.

Here, “temporally controlling a workload” refers to such control that a workload is temporarily delayed in execution within the same DC base. In this way, by delaying many workloads from a time range where renewable energy supply is insufficient to a time range where renewable energy supply is surplus, it is possible to increase the renewable energy utilization ratio.

On the other hand, “spatially controlling a workload” refers to such control that a location where a workload is executed is moved from a certain DC base to another DC base. In this way, as in the case of temporal control, by moving the workload from a DC base where renewable energy supply is insufficient to another DC base where renewable energy supply is abundant, it is possible to increase the renewable energy utilization ratio.

Generally, workloads executed at DC bases can be broadly divided into two types: interactive workload and batch job. An interactive workload is one such as a Web application that needs to be processed in real time and is basically always in a running condition. A batch job is one such as image processing or machine learning training process that does not necessarily need to be processed immediately, and may be executed by a predetermined time and processing thereof can be delayed within such a range.

Although the interactive workload here can be spatially controlled, processing of the interactive workload cannot be delayed, and therefore the interactive workload is not suitable for temporal control. On the other hand, processing of a batch job can be delayed, and so it can be temporally controlled. Thus, for clarity, the former will be referred to as a “spatial-migration-oriented workload” and the latter will be referred to as a “temporal-migration-oriented workload.”

Japanese Patent Laid-Open No. 2021-189845 discloses a technology to efficiently utilize renewable energy generated at DC bases by migrating workload execution process from a DC base where renewable energy supply is predicted to be insufficient to a DC base where renewable energy supply is predicted to be surplus.

Using the technology in Japanese Patent Laid-Open No. 2021-189845 makes it possible to perform spatial control of workloads according to renewable energy supply at the DC bases and adjust the ratio of workloads executed at the DC bases. This allows the renewable energy utilization ratio to be improved at the DC bases.

However, workloads executed at the DC bases cannot always migrate between the DC bases freely. When workloads migrate between the DC bases, downtimes may be produced on no small number of workloads, and so users who execute the workloads may limit time ranges in which the workloads can migrate between the DC bases. For example when a workload is something like a Web application, trouble may occur if a downtime is produced in a time range frequently accessed by users, and so spatial migration of such workloads may be set to be non-migratable. In such a time range, since it is not possible to spatially control workloads, DC power demand may not be adjustable.

An object of the present invention, which has been made in view of such backgrounds, is to provide a data center system, a DC inter-base workload control method and a DC inter-base workload control system that can perform spatial control of workloads between DC bases and temporal control within a DC base while taking into account time ranges in which no workload is migratable to thereby adjust DC power demand so as to improve the renewable energy utilization ratio at all the DC bases.

SUMMARY

As an aspect of solving the above-described problems, the invention provides a data center system including a plurality of bases provided with a calculation resource and a renewable energy power supply to execute a workload, and an inter-base workload control system to control migration of the workload between the plurality of bases and within a same base, in which a processor of the inter-base workload control system manages a positive excess power amount obtained by subtracting a predicted amount of power supply of the renewable energy power supply from a predicted amount of power consumption of the calculation resource associated with execution of the workload in a future time range at the bases, spatial migratable time range information

on a spatial migratable time range where spatial migration is possible and the workload scheduled to be executed in the future time range at the bases can be migrated to another base and a non-migratable time range where such migration is not possible, temporal migratable time range information on temporal migration whereby the workload scheduled to be executed in the future time range at the bases is delayed in execution within the same base to be migrated to another time range and a predicted amount of power consumption which is a predicted amount of power consumption of the calculation resource associated with execution of the workload scheduled to be executed in the future time range at the bases, determines a power adjustment amount indicating the predicted amount of power consumption that moves through the spatial migration and the temporal migration of the workload scheduled to be executed in the future time range at the bases based on the spatial migratable time range information, the temporal migratable time range information and the predicted amount of power consumption so that a sum total of the excess power amounts in the future time range at the plurality of bases becomes smaller, and executes the spatial migration and the temporal migration of the workload based on the power adjustment amount.

According to the present invention, it is possible to perform spatial control of workloads between DC bases and temporal control within a DC base while taking into account a workload non-migratable time range to thereby adjust DC power demand so as to improve a renewable energy utilization ratio at all DC bases.

The details of one or more implementations of the subject matter described in the specification are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages of the subject matter will become apparent from the description, the drawings, and the claims.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a diagram illustrating an example of a configuration of a data center system according to the present embodiment;

FIG. 2 is a diagram illustrating an example of a configuration of a base management computer;

FIG. 3 is a diagram illustrating an example of a configuration of an inter-base workload control system;

FIG. 4 is a diagram illustrating an example of a renewable energy power table;

FIG. 5 is a diagram illustrating an example of a power demand table;

FIG. 6 is a diagram illustrating an example of a workload table;

FIG. 7 is a diagram illustrating an example of a spatial power migration table;

FIG. 8 is a diagram illustrating an example of a supply/demand adjustment amount ratio table;

FIG. 9 is a flowchart describing an overview of a workload control planning process;

FIG. 10 is a flowchart describing details of a base-by-base data update process;

FIG. 11 is a flowchart describing a workload control execution process;

FIG. 12 is a flowchart describing details of a workload relocation planning process;

FIG. 13 is a flowchart describing details of a power determination process of temporal migration; and

FIG. 14 is a diagram illustrating an example of a workload input screen.

DESCRIPTION OF EMBODIMENTS

Hereinafter, embodiments of the present invention will be described in detail based on the accompanying drawings. The present embodiments are just an example of how to implement the present invention and are not limiting the technical scope of the present invention. Various elements described in the present embodiments and a combination of all those elements are not necessarily essential to the means for providing solutions to the problems of the invention.

In the following descriptions, identical reference numerals are assigned to identical components in principle and repetitive descriptions are omitted. The number of respective components in the following descriptions are not limited unless otherwise noted. In the following descriptions, when identical components are described in a distinct manner, reference numerals including subscripts are used, whereas when identical components are described without distinction, reference numerals excluding the subscripts are used.

In the following descriptions, there are cases where processes executed by a program may be described. A computer performs processes defined by the program using a processor (e.g., CPU (Central Processing Unit), GPU (Graphics Processing Unit)) while using a memory of a main storage apparatus or the like. Therefore, the entity of processes performed by executing the program may be regarded as the processor. By the processor executing the program, functional parts that perform the processes are implemented.

Similarly, the entity of processes performed by executing the program may be a controller, an apparatus, a system, a computer or a node, each having a processor. The entity of processes performed by executing the program needs only to be a computation section or may include a dedicated circuit that performs specific processing. Examples of the dedicated circuit include FPGA (Field-Programmable Gate Array) and ASIC (Application Specific Integrated Circuit).

In the following descriptions, the program may be installed in the computer from a program source. The program source may be, for example, a program distribution server or computer-readable non-transitory storage media. In the case where the program source is a program distribution server, the program distribution server includes a processor and a storage resource (storage) that stores programs to be distributed and the processor of the program distribution server may distribute the programs to be distributed to other computers. Moreover, in the present embodiments, two or more programs may be implemented as one program or one program may be implemented as two or more programs.

In the following descriptions, various kinds of data will be described in table formats. However, the data formats are not limited to table formats, but may be other data formats such as a cue, a list or a CSV (Comma Separated Value).

In the following descriptions, IDs (identifiers), numbers or the like are used as information to identify some targets, but without being limited to them, identification information of various kinds may be used.

FIG. 1 is a diagram illustrating an example of a configuration of a data center system S according to the present embodiment.

The data center system S is composed of one or a plurality of DC (Data Center) bases **1000** and an inter-base workload control system **10000**. The DC bases **1000** are communicatively coupled by a wide area network **9000**.

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The DC base **1000** is provided with a base management computer **2000**, one or a plurality of servers **3000** used by an administrator or a user of the DC base **1000** and one or a plurality of storages **4000** used by the administrator or the user of the DC base **1000**. The servers **3000** and the storages **4000** are communicatively coupled by a data network **6000**. Furthermore, the base management computer **2000**, the servers **3000** and the storages **4000** are communicatively coupled by a management network **5000**.

Note that the management network **5000**, the data network **6000** and the wide area network **9000** are wired or wireless communication networks such as internet, LAN (Local Area Network), WAN (Wide Area Network) or dedicated line.

The power for driving the base management computer **2000**, the servers **3000** and the storages **4000** is supplied from a power busbar **8000** within the DC bases **1000** and can be supplied by a renewable power supply **7000** within the DC base **1000** or a system power supply **11000** from outside. The renewable power supply **7000** is power generation equipment of renewable energy such as sunlight, wind power installed in the DC bases **1000** and the power generated within the DC base **1000** can be used within the DC base **1000**.

The servers **3000** and the storages **4000** execute various kinds of processing such as the aforementioned spatial-migration-oriented workload or temporal-migration-oriented workload.

The inter-base workload control system **10000** allocates a processing load (workload) for the system, execution of which is instructed by the user to each DC base **1000**. The inter-base workload control system **10000** also causes a workload already executed at a DC base to migrate to another DC base as appropriate.

The base management computer **2000** manages workloads allocated by the inter-base workload control system **10000** to each DC base **1000** in predetermined time intervals (time slots) and controls timing at which the workload is executed within each DC base.

Note that in the present embodiments, a workload may refer to a job (process) itself. In the following descriptions, the interval of a time slot is supposed to be one hour, but the time slot interval is not limited to such an interval, and any given time interval can be set.

FIG. 2 is a diagram illustrating an example of a configuration of the base management computer **2000**.

The base management computer **2000** is provided with a storage apparatus **12000** such as an HDD (Hard Disk Drive) or an SSD (Solid State Drive). The base management computer **2000** is provided with a processing apparatus **13000** (processor) such as a CPU (Central Processing Unit), a DSP (Digital Signal Processor), a GPU (Graphics Processing Unit) or an FPGA. The base management computer **2000** is provided with a main storage apparatus **14000** (memory) such as a ROM (Read Only Memory) or a RAM (Random Access Memory). The base management computer **2000** is provided with a communication apparatus **15000** composed of a NIC (Network Interface Card), a wireless communication module, a USB (Universal Serial Bus) (registered trademark, the same applies hereafter) module or a serial communication module. The base management computer **2000** is provided with an input apparatus **16000** composed of a mouse and a keyboard or the like and an output apparatus **17000** composed of a liquid crystal display, an organic EL (Electro-Luminescence) display or the like.

The base management computer **2000** stores various programs such as an intra-base workload control program

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12500, a power demand prediction program **12600** and a renewable energy prediction program **12700**.

The intra-base workload control program **12500** calculates a target amount of power consumption in a future time slot so as to maximize a renewable energy utilization ratio based on a predicted amount of power consumption in the future time slot at the DC base **1000**. Based on the determined target amount of power consumption, the intra-base workload control program **12500** determines execution timing of each workload in the future time slot and executes each workload at the determined timing.

The power demand prediction program **12600** calculates a predicted amount of power consumed at the DC base **1000** in the future time slot.

The renewable energy prediction program **12700** calculates a predicted amount of renewable energy power supplied to the DC base **1000** in the future time slot. The renewable energy supplied to the DC base **1000** may be generated by the renewable power supply **7000** or may be supplied according to a PPA (Power Purchase Agreement) or the like.

Furthermore, the base management computer **2000** stores respective databases of a renewable energy power table **12100**, a power demand table **12200**, a workload table **12300** and a spatial power migration table **12400**.

The renewable energy power table **12100** stores a predicted amount of renewable energy to be supplied, predicted by the renewable energy prediction program **12700** in each time slot and its actual value.

The power demand table **12200** stores a predicted amount and an actual amount of power consumed at the DC bases **1000** in each time slot and a power consumption target amount.

The workload table **12300** stores and accumulates predicted values and actual values of power consumption of workloads executed at the respective DC bases **1000** and information on execution schedules, spatial migratable time and spatial non-migratable time. The servers **3000** and the storages **4000** execute each workload according to the workload table **12300**.

The spatial power migration table **12400** stores information on an amount of power of a spatial-migration-oriented workload to be relocated from a certain DC base **1000** to another DC base **1000** in each time slot.

FIG. 3 is a diagram illustrating an example of a configuration of the inter-base workload control system **10000**.

The inter-base workload control system **10000** is provided with a storage apparatus **18000** such as an HDD, an SSD, a processing apparatus **19000** (processor) such as a CPU, a DSP, a GPU or an FPGA and a main storage apparatus **20000** (memory) such as a ROM or a RAM. The inter-base workload control system **10000** is provided with a communication apparatus **21000** composed of a NIC, a wireless communication module, a USB module, a serial communication module or the like. The inter-base workload control system **10000** is provided with an input apparatus **22000** composed of a mouse, a keyboard or the like and an output apparatus **23000** composed of a liquid crystal display, an organic EL display or the like.

The inter-base workload control system **10000** stores respective programs of an inter-base initial location program **18300**, an inter-base relocation program **18400** and an inter-base data collection program **18500**.

The inter-base initial location program **18300** calculates a ratio at which initial location of a workload should be performed at the DC bases in a future time slot based on an amount of renewable energy to be supplied and power

demand in the future time slot at the DC bases **1000**. The inter-base initial location program **18300** then performs control to allocate a workload to each DC base **1000** based on the calculated ratio at the clock time.

The inter-base relocation program **18400** determines what amount of power of a spatial-migration-oriented workload already executed at a certain DC base should be relocated to another DC base in the future time slot. This amount of power is determined based on information on an amount of renewable energy to be supplied, power demand, a spatial migratable time range and a spatial non-migratable time range in the future time slot at the DC bases **1000**. The inter-base relocation program **18400** then performs workload relocation at the clock time in the future time slot.

The inter-base data collection program **18500** collects information on the power demand and the amount of renewable energy to be supplied, predicted by the power demand prediction program **12600** and the renewable energy prediction program **12700** respectively, at the DC bases **1000**.

Furthermore, the inter-base workload control system **10000** stores respective databases of a base-by-base data **18100** and a supply/demand adjustment amount ratio table **18200**.

The base-by-base data **18100** stores information on the power demand and the amount of renewable energy to be supplied or the like, collected from each DC base.

The supply/demand adjustment amount ratio table **18200** stores a supply/demand adjustment amount ratio that indicates at which DC base and in what ratio workloads should be initially located in each time slot.

Next, specific examples of the respective databases will be described.

(Renewable Energy Power Table)

FIG. 4 is a diagram illustrating an example of the renewable energy power table **12100** for each DC base **1000**. The renewable energy power table **12100** includes data items of a time slot ID **12110** in which information on time slot identification is set and clock time **12120** in which the time slot start clock time is set. The renewable energy power table **12100** includes data items of a renewable energy supply amount prediction **12130** in which a predicted amount of renewable energy to be supplied at the target clock time is set. The renewable energy power table **12100** includes data items of a renewable energy supply amount actual measurement **12140** in which a measured amount of renewable energy to be supplied, actually measured at the target clock time is set. The renewable energy power table **12100** is composed of one or more records having these data items.

The renewable energy supply amount prediction **12130** includes an amount of power supplied from a renewable energy power supply (renewable power supply **7000**) in the future time range (time slot) at the DC bases **1000** and a predicted amount of power supplied or purchased from outside according to the PPA or the like.

Note that each predicted value and measured value in the renewable energy power table **12100** may be inputted by the user or automatically acquired from a predetermined database.

(Power Demand Table)

FIG. 5 is a diagram illustrating an example of the power demand table **12200** for each DC base **1000**. The power demand table **12200** includes data items of a time slot ID **12210** in which time slot identification information is set and clock time **12220** in which the time slot start clock time is set. The power demand table **12200** includes data items of a DC power consumption amount prediction **12230** in which a predicted amount of power consumption at the DC base

1000 at the target clock time is set. The power demand table **12200** includes data items of a DC power consumption amount actual measurement **12240** in which a measured amount of power consumption at the DC base **1000** actually measured at the target clock time is set. The power demand table **12200** includes data items of a power consumption target value **12250** indicating a target amount of power when the intra-base workload control program **12500** actually deploys a workload at the target clock time. The power demand table **12200** is composed of one or more records having these data items.

The DC power consumption amount prediction **12230** is a predicted amount of power consumption including a calculation resource (server **3000**, storage **4000**) associated with workload execution in a future time range (time slot) at the DC bases **1000** and network equipment (not shown), air conditioning or the like.

Note that the respective predicted values and measured values in the power demand table **12200** may be inputted by the user or may be automatically acquired from a predetermined database.

(Workload Table)

FIG. 6 is a diagram illustrating an example of the workload table **12300**. The workload table describes information on workloads allocated through the inter-base workload control system **10000** at the DC bases **1000**. The workload table **12300** includes data items of a workload ID **12305** in which workload identification information is set and a power consumption amount prediction **12310** in which a predicted amount of power consumption in a workload is set. The workload table **12300** includes data items of a power consumption amount actual measurement **12315** in which a measured amount of power consumption in a workload is set and input time **12320** in which the time at which workload information is inputted to the base management computer **2000** is set. Furthermore, in the case of a temporal-migration-oriented workload, the workload table **12300** includes data items of an execution schedule **12325** representing the clock time at which workload execution may be started. The workload table **12300** includes data items of a changed execution schedule **12330** in which execution timing changed (delayed) by the intra-base workload control program **12500** from the execution schedule **12325** regarding the workload is set. The workload table **12300** includes data items of a delayable time **12335** representing the delayable time when the workload is a temporal-migration-oriented workload and an executable base **12340** in which the DC base **1000** that can execute a workload is set. Regarding a temporal-migration-oriented workload, when the delayable time **12335** is 0 or when the intra-base workload control program **12500** determines that there is no need for the delay, the changed execution schedule **12330** becomes n/a, and the workload is executed at the clock time according to the execution schedule **12325**. When enough time is set as the delayable time **12335** and when the intra-base workload control program **12500** determines that the workload will be delayed, the clock time is written in the changed execution schedule **12330** and the workload is executed at that clock time. The workload table **12300** includes data items of the spatial migratable time range **12345** in which a spatial migratable time range is set when the workload is a spatial-migration-oriented workload. The workload table **12300** includes data items of the spatial non-migratable time range **12350** in which a spatial non-migratable time range is set when the workload is a spatial-migration-oriented workload. The workload table **12300** includes data items of an execution status **12355** in which a workload execution status is set.

The workload table **12300** is composed of one or more records having these data items.

The delayable time **12335** is temporal migratable time information on temporal migration whereby a workload scheduled to be executed in a future time range (time slot) at the DC bases is delayed in execution within the same DC base **1000** and migrated to another time range (time slot).

The spatial migratable time range **12345** is information on a spatial migratable time range where a workload scheduled to be executed in a future time range (time slot) at a DC base **1000** is migrated to another DC base **1000**. The spatial non-migratable time range **12350** is information on a non-migratable time range where it is not possible to migrate a workload scheduled to be executed in a future time range (time slot) at a DC base **1000** to another DC base **1000**. The spatial migratable time range **12345** and the spatial non-migratable time range **12350** are information on the spatial migratable time range.

In the case of a temporal-migration-oriented workload, the user inputs the workload after writing the execution schedule **12325**, the delayable time **12335** representing maximum time by which workload execution can be delayed from the execution schedule **12325** and the executable base **12340**. In the case of a spatial-migration-oriented workload, the user inputs the workload after writing the spatial migratable time range **12345** and the spatial non-migratable time range **12350**. In the case of the spatial-migration-oriented workload, the workload is executed immediately at the clock time the workload is inputted, and so no information is held on the execution schedule **12325** and the delayable time **12335**.

(Spatial Power Migration Table)

FIG. 7 is a diagram illustrating an example of the spatial power migration table **12400**. The spatial power migration table **12400** includes data items of a time slot ID **12410** in which time slot identification information is set and clock time **12420** in which the time slot start clock time is set. The spatial power migration table **12400** includes data items of a DC base (N) **12430** representing a power amount of spatial-migration-oriented workload to be relocated to a DC base **1000** other than the own base at the target clock time. The spatial power migration table **12400** is composed of one or more records having these data items.

The number of columns saved in the DC base (N) **12430** is equal to the number of other DC bases **1000** except the DC base **1000** itself in which this table is saved. For example, if the number of DC bases **1000** is assumed to be M, the number of columns saved in the DC base (N) **12430** is M-1. (Supply/Demand Adjustment Amount Ratio Table)

FIG. 8 is a diagram illustrating an example of the supply/demand adjustment amount ratio table **18200**. The supply/demand adjustment amount ratio table **18200** includes data items of a time slot ID **18210** in which time slot identification information is set and clock time **18220** in which the time slot start clock time is set. The supply/demand adjustment amount ratio table **18200** includes data items of a DC base (N) **18230** representing at what ratio workloads inputted to the inter-base workload control system **10000** are allocated to the respective DC bases **1000** at the target clock time. The supply/demand adjustment amount ratio table **18200** is composed of one or more records having these data items.

The number of columns saved in the DC base (N) **18230** is equal to the number of DC bases **1000**. For example, if the number of DC bases **1000** is assumed to be M, the number of columns saved in the DC base (N) **18230** is M.

If the above-described programs are executed by the base management computer **2000**, the programs are executed by the processing apparatus **13000** reading (the programs stored in the main storage apparatus **14000** or the storage apparatus **12000**). Similarly, if the programs are executed by the inter-base workload control system **10000**, the programs are executed by the processing apparatus **19000** reading (the program stored in the main storage apparatus **20000** or the storage apparatus **18000**). Each program can be recorded in a recording medium and distributed.

Note that all or part of the base management computer **2000** and the inter-base workload control system **10000** may be implemented using virtual information processing resources supplied using a virtualization technology or a process space separation technology or the like. It can be, for example, a virtual server supplied by a cloud system. All or part of functions provided by the base management computer **2000** and the inter-base workload control system **10000** may be implemented by a service provided by a cloud system via an API (Application Programming Interface).

Next, processes executed by the base management computer **2000** and the inter-base workload control system **10000** will be described.

<Workload Control Planning Process>

FIG. 9 is a flowchart describing an overview of a workload control planning process, which is a process of planning in advance how the DC base **1000** controls a workload in a future time slot. The workload control is executed by repeating the workload control planning process (FIG. 9) and a workload control execution process (FIG. 11), which will be described later.

First of all, the workload control planning process is repetitively executed in a preliminary stage of actually controlling a workload every predetermined clock time (e.g., daily), at a predetermined time interval (e.g., predetermined hours and minutes before each time slot starts) or at predetermined timing (user-specified clock time). For example, in a stage of the day before actually controlling a workload, how to control the workload on the next day is planned. However, at the stage of the day before, it is impossible to exactly know what kind of workload is inputted on the day of control. Therefore, instead of making a control plan about individual workloads, a rough policy can be made about how to control an approximate amount of power or a ratio of the amount of power predicted to be consumed by the workload in each future time slot. The "control" is temporal migration or spatial migration.

The inter-base workload control system **10000** plans a supply/demand adjustment amount ratio indicating the ratio of how much of the inputted workload should be allocated in each future time slot. A plan is made as to what amount of power consumed by a spatial-control-oriented workload already executed at a certain DC base **1000** should be migrated to another DC base **1000**.

The base management computer **2000** plans how to temporally control the power consumed by the temporal-migration-oriented workload within the DC base **1000** in each future time slot.

After that, in the workload control execution process, individual workload control is performed based on spatial control and temporal control policies (what amount of power should be temporally or spatially migrated) actually planned in the workload control planning process at the clock time in the time slot.

First in step S1000, the base management computer **2000** executes a base-by-base data update process of predicting an amount of renewable energy to be supplied at the DC bases

1000 and an amount of DC power consumption and accumulating these past data. Details of the base-by-base data update process (S1000) will be described later.

Next, in step S2000, the inter-base data collection program 18500 in the inter-base workload control system 10000 collects update data and stores the update data in the base-by-base data 18100. The update data is the amount of renewable energy to be supplied at all the DC bases 1000 updated in step S1000, the predicted amount of DC power consumption and the actual amount.

Next, in step S3000, the inter-base initial location program 18300 in the inter-base workload control system 10000 calculates a supply/demand adjustment amount ratio in the future time slot based on information on each DC base 1000 collected in step S2000. The supply/demand adjustment amount ratio is stored at the DC base (N) 18230 in the supply/demand adjustment amount ratio table 18200. As described, for example, in Japanese Patent Laid-Open No. 2021-189845, the method for calculating the supply/demand adjustment amount ratio may be calculated based on a supply/demand balance between the amount of renewable energy to be supplied and the amount of power consumption at the DC bases 1000 or using other methods.

At the DC bases 1000, the base management computer 2000 calculates a target amount of power consumption in the future time slot based on the supply/demand adjustment amount ratio, the amount of renewable energy to be supplied at the base and the predicted amount of DC power consumption determined by the inter-base workload control system 10000. The target amount of power consumption is stored in the power consumption target value 12250 in the power demand table 12200. Any given method may be used as the method for calculating the target amount of power consumption as long as it is a method for determining an amount of power that improves the renewable energy utilization ratio.

Next, in step S4000, the inter-base relocation program 18400 of the inter-base workload control system 10000 executes a workload relocation planning process. In the workload relocation planning process, each DC base 1000 in the future time slot makes a plan of relocating an already executed workload at another DC base 1000. Details of the workload relocation planning process (S4000) will be described later.

Next, in step S5000, a control parameter of each DC base 1000 updated by the inter-base workload control system 10000 in step S4000 is synchronized with that of each DC base 1000. The above-described processes are repetitively executed.

<Base-by-Base Data Update Process>

Next, details of a base-by-base data update process (S1000) will be described. FIG. 10 is a flowchart describing the base-by-base data update process (S1000). This process is executed at all the DC bases 1000.

First, in step S1100, the renewable energy prediction program 12700 predicts an amount of renewable energy to be supplied in respective time slots after the present. More specifically, for example, the renewable energy prediction program 12700 acquires each value of the clock time 12120 corresponding to each record of the renewable energy power table 12100 and each value of the renewable energy supply amount actual measurement 12140. The renewable energy prediction program 12700 predicts an amount of renewable energy to be supplied in a time slot after the present based on a predetermined algorithm (e.g., by executing a time sequence analysis or creating a prediction model through machine learning or the like) for the acquired values. The

renewable energy prediction program 12700 stores each predicted amount of supply in the renewable energy supply amount prediction 12130 of each time slot record of the renewable energy power table 12100.

Note that the data used by the renewable energy prediction program 12700 to predict an amount of renewable energy to be supplied is not limited to the measured value of renewable energy supplied as shown here. In other words, the data may be used in combination with other information such as meteorological data at the DC base 1000.

The renewable energy prediction program 12700 acquires amounts of renewable energy supplied in past time slots from a predetermined apparatus (e.g., from an external database or server). The renewable energy prediction program 12700 stores the acquired amount of renewable energy supplied in the renewable energy supply amount actual measurement 12140 of records associated with the time slots of the renewable energy power table 12100 as actual values.

Next, in step S1200, the power demand prediction program 12600 predicts an amount of power consumption in each time slot after the present. More specifically, for example, the power demand prediction program 12600 acquires the respective values of the respective records at the clock time 12220 and the DC power consumption amount actual measurement 12240 in the power demand table 12200. The power demand prediction program 12600 predicts an amount of power consumption for all the DC bases based on a predetermined algorithm (e.g., by executing a time sequence analysis or creating a prediction model through machine learning) in each time slot after the present for the acquired value. The power demand prediction program 12600 stores the respective predicted amounts of power consumption in the DC power consumption amount prediction 12230 of each time slot record in the power demand table 12200.

Note that the data used by the power demand prediction program 12600 for prediction of the amount of power consumption is not limited to the measured values of power consumption shown here, but the data may be used in combination with other information such as meteorological data in the DC base 1000.

The power demand prediction program 12600 acquires an amount of power consumption of the entire DC base at the DC base 1000 in a past time slot from predetermined apparatuses (e.g., external databases or servers). The power demand prediction program 12600 stores the acquired amount of power consumption in the DC power consumption amount actual measurement 12240 of a record associated with the time slot in the power demand table 12200 as an actual value.

<Workload Control Execution Process>

The workload control execution process is a process of determining at which DC base 1000 and when individual workloads should be executed based on the control policies in each time slot planned in the workload control planning process. The workload control execution process is composed of two processes: a workload initial location control process and a workload relocation control process. The workload initial location control process is a control process for a workload before execution and the workload relocation control process is a control process for a workload already executed at any DC base 1000.

FIG. 11 is a flowchart describing the workload control execution process.

First, in step S6000, the inter-base initial location program 18300 and the intra-base workload control program 12500 perform the workload initial location control process.

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When there is a workload that the user wants to execute at any one of the DC bases **1000**, the user provides an instruction for inputting a workload to the inter-base workload control system **10000**. At this time, the user specifies the DC base **1000** that is allowed to execute the workload by taking into account characteristics of the workload scheduled to be executed. For example, since only the DC bases **1** and **2** satisfy an SLA (Service Level Agreement) such as latency, the user may allow locations at the DC bases **1** and **2**, but the user may not allow locations at the other DC bases **1000**. When the workload is a temporal-migration-oriented workload, the user specifies information about how much delay from input to execution is allowed.

The inter-base initial location program **18300** determines at which DC base **1000** to locate the workload based on the supply/demand adjustment amount ratio **18230** calculated in step **S3000** of the workload control planning process and the information on at which DC base **1000** the workload can be located.

The workload allocated to any one DC base **1000** by the inter-base initial location program **18300** is immediately executed if it is a spatial-control-oriented workload. If it is a temporal-control-oriented workload, the intra-base workload control program **12500** determines the appropriate execution time for the temporal-control-oriented workload based on the power consumption amount prediction **12310** of the workload and stores the appropriate execution clock time in the changed execution schedule **12330**. In such a case, considering the information of the delayable time **12335**, the program ensures that the power consumption amount in each time slot is as close to the power consumption target value **12250** as possible. After that, when the clock time set in the changed execution schedule **12330** comes, the program executes the workload.

The location control of the DC base **1000** performed by the inter-base initial location program **18300** and the temporal control performed by the intra-base workload control program **12500** together are called a “workload initial location control process.”

Next, in step **S7000**, the inter-base relocation program **18400** executes a workload relocation control process. The inter-base relocation program **18400** determines a workload to migrate to each DC base **1000** based on the power consumption amount prediction **12310** of the workload ensuring that the amount of power is as close to the amount of power stored at the DC base (N) **12430** in the spatial power migration table **12400** as possible. Then, the inter-base relocation program **18400** executes actual migration. <Workload Relocation Planning Process>

FIG. 12 is a flowchart describing details of the workload relocation planning process (**S4000**).

First, in step **S4110**, the inter-base relocation program **18400** selects the first one out of the DC bases **1000**. Any one DC base **1000** can be selected.

Next, in step **S4120**, the inter-base relocation program **18400** selects the most recent time slot. More specifically, the inter-base relocation program **18400** selects a record of the clock time **12120** indicating the most recent future clock time from the current clock time with reference to the renewable energy power table **12100**.

Next, in step **S4130**, the inter-base relocation program **18400** determines whether or not there is any power excess in a time slot currently targeted. The power excess (amount of excess power) is a positive value obtained by subtracting a predicted amount of a renewable energy power supply (renewable power supply **7000**) from a predicted amount of power consumption including calculation resources associ-

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ated with workload execution in a future time range (time slot) at the DC bases **1000**, network equipment (not shown), air conditioning or the like. The calculation resources here are the servers **3000** and the storages **4000**.

More specifically, it is determined in the time slot currently targeted whether the value (the power consumption target value **12250**—the renewable energy supply amount prediction **12130**) is 0 or more. If (the power consumption target value **12250**—the renewable energy supply amount prediction **12130**) is 0 or more, the inter-base relocation program **18400** determines that there is power excess and proceeds to step **S4140**. Otherwise, the inter-base relocation program **18400** determines that there is no power excess and proceeds to step **S4160**.

In step **S4140**, the inter-base relocation program **18400** determines the power to be spatially relocated in the time slot currently targeted so that the power may be smaller than the sum total of power excess at all the DC bases **1000** in the data center system **S** as a whole. The inter-base relocation program **18400** preferably ensures that the sum total is minimized. The inter-base relocation program **18400** determines the power for migration to another DC base **1000** from the DC base **1000** currently targeted, based on the amount of renewable energy supplied at the DC bases **1000** collected in step **S2000** of the workload control planning process, the power consumption amount data, the spatial migratable time range **12345** of the spatial-migration-oriented workload and the spatial non-migratable time range **12350**. The number of migration destination DC bases **1000** need not be singular, but can be plural.

In step **S4150**, the inter-base relocation program **18400** decreases the DC power consumption amount prediction **12230** in the power demand table **12200** at the DC base **1000** currently targeted and the power consumption target value **12250** by the power amount corresponding to the migration in step **S4140**. On the other hand, the inter-base relocation program **18400** increases the DC power consumption amount prediction **12230** in the power demand table **12200** at the DC base **1000**, the migration destination and the power consumption target value **12250** by the power amount corresponding to the migration.

In step **S4160**, the inter-base relocation program **18400** determines whether or not the time slot currently targeted is the last time slot. If the time slot is the last one, the inter-base relocation program **18400** proceeds to step **S4210**; otherwise it proceeds to step **S4170**.

In the workload control planning process, the user determines in advance how far into the future time slots the plan will cover, which is saved in the storage apparatus **18000** of the inter-base workload control system **10000**. For example, if a workload control plan for the next one day is made, the last time slot corresponds to a time slot after 24 hours.

In step **S4170**, the inter-base relocation program **18400** determines over again whether or not there is power excess in the time slot currently targeted in the same manner as step **S4130**. If the existence of power excess is determined, the inter-base relocation program **18400** proceeds to step **S4180**; otherwise it proceeds to step **S4200**.

In step **S4180**, the inter-base relocation program **18400** determines whether or not there is any spatial non-migratable workload in the time slot currently targeted. More specifically, the inter-base relocation program **18400** determines whether or not there is any workload in which the execution status **12355** in the workload table **12300** indicates “execution in progress” and in which the time set in the spatial non-migratable time range **12350** includes the clock time of the time slot currently targeted. If there is such a

workload, the inter-base relocation program **18400** proceeds to step **S4190**; otherwise it proceeds to step **S4200**.

In step **S4200**, the inter-base relocation program **18400** targets the time slot at the clock time next to the time slot currently targeted and proceeds to step **S4130**.

In step **S4210**, the inter-base relocation program **18400** determines whether or not all the DC bases **1000** have been targeted. If all the DC bases **1000** have been targeted, the inter-base relocation program **18400** terminates the workload relocation planning process. If there is any DC base **1000** that remains untargeted, the inter-base relocation program **18400** proceeds to step **S4220**.

In step **S4220**, the inter-base relocation program **18400** selects one out of the non-targeted DC bases **1000** as appropriate and makes it the next target. After that, the inter-base relocation program **18400** proceeds to step **S4120**. <Temporal Migration Power Amount Determination Process>

FIG. **13** is a flowchart describing details of a temporal migration power amount determination process (**S4190**).

First, in step **S4310**, the inter-base relocation program **18400** targets a time slot next to the time slot currently targeted. The time slot targeted in the temporal migration power amount determination process (**S4190**) is held independently of the time slot targeted in the workload relocation planning process (**S4000**). For example, suppose that a certain time slot is targeted in the workload relocation planning process (**S4000**) and the process in step **S4310** is executed for the first time in the temporal migration power amount determination process (**S4190**). In this case, the inter-base relocation program **18400** targets the time slot next to the clock time of the time slot targeted in the workload relocation planning process (**S4000**).

However, that never causes the time slot itself held in the inter-base relocation program **18400** to be changed in the workload relocation planning process (**S4000**). In the temporal migration power amount determination process (**S4190**), if step **S4310** is executed from the second time onward, the inter-base relocation program **18400** targets a time slot next to the previously targeted clock time.

In the description in FIG. **13**, the “time slot currently targeted” shall refer to the time slot targeted in the temporal migration power amount determination process (**S4190**). In other words, the “time slot currently targeted” refers to one different from the time slot targeted in the workload relocation planning process (**S4000**).

Next, in step **S4320**, the inter-base relocation program **18400** acquires a workload that is changed from a spatial non-migratable time to a spatial migratable time in the time slot currently targeted. More specifically, the inter-base relocation program **18400** refers to the execution status **12355**, the spatial migratable time range **12345** and the spatial non-migratable time range **12350** in the workload table **12300**. The inter-base relocation program **18400** acquires records of a spatial-migration-oriented workload showing that the execution status **12355** is execution in progress, the time slot currently targeted is included in the spatial migratable time range **12345** and the preceding time slot is included in the spatial non-migratable time range **12350**.

In step **S4330**, the inter-base relocation program **18400** determines whether or not the spatial-migration-oriented workload acquired in step **S4320** can migrate to another DC base **1000**. The inter-base relocation program **18400** determines whether or not the DC base **1000**, the migration destination, is found in the spatial-migration-oriented workload acquired in step **S4320** from the renewable supply

amount, the amount of power consumption and the target amount of power consumption of the DC base **1000** or the like saved in the base-by-base data **18100**. If even one workload has the migration destination, the inter-base relocation program **18400** proceeds to step **S4340**; otherwise, it proceeds to step **S4360**.

In step **S4340**, the inter-base relocation program **18400** temporally migrates to the time slot from the time slot targeted by the inter-base relocation program **18400** in the workload relocation planning process (**S4000**). Here, the power amount to be temporally migrated is a power amount up to the power amount of the spatial-migration-oriented workload that spatially migrates to another DC base **1000** in step **S4330**. The inter-base relocation program **18400** subtracts the power consumption target value **12250** in the power demand table **12200** by the power amount in the time slot targeted in the workload relocation planning process (**S4000**).

In other words, when the power excess (excess power amount) assumed to still exist in step **S4170** is equal to or smaller than a second power adjustment amount migrating through spatial migration of the workload acquired in step **S4320**, a first power adjustment amount by which to delay is determined in step **S4340**. Due to the delay by the first power adjustment amount, part or all of the power excess in the corresponding time range moves to the next time range up to the second power adjustment amount.

When the power excess (excess power amount) assumed to still exist in step **S4170** exceeds the second power adjustment amount that moves through spatial migration of the workload acquired in step **S4320**, the first power adjustment amount by which to delay is determined in step **S4340**. Due to the delay by the first power adjustment amount, part of the power excess in the corresponding time range moves to the next time range up to the second power adjustment amount.

In the process in step **S4340**, a temporal-migration-oriented workload is migrated to the time slot to be currently targeted in the workload relocation planning process (**S4000**) from the time range in the targeted time slot in the temporal migration power amount determination process (**S4180**). In the time slot currently targeted in the temporal migration power amount determination process (**S4180**), a spatial-migration-oriented workload having a power amount identical or similar to the temporally migrated power amount is migrated to another DC base **1000**. Thus, the power amount decreases in the time slot targeted in the workload relocation planning process (**S4000**), whereas there is no increase or decrease in the time slot currently targeted in the temporal migration power amount determination process (**S4180**). This is reflected in the power consumption target value **12250** in the power demand table **12200**.

In step **S4350**, the inter-base relocation program **18400** determines whether or not there is any power excess in the time slot targeted in the workload relocation planning process (**S4000**). This determination is made using a method similar to the method in step **S4130** in the workload relocation planning process. When there is power excess, the inter-base relocation program **18400** proceeds to step **S4360**; otherwise, it terminates the process.

In step **S4360**, the inter-base relocation program **18400** determines whether or not the time slot currently targeted is the last time slot. This determination is made using a method similar to the method in step **S4190** in the workload relocation planning process. If the targeted time slot is the last one, temporal control is not possible after this time slot, and

so the inter-base relocation program **18400** terminates the temporal migration power amount determination process. On the other hand, the inter-base relocation program **18400** proceeds to step **S4310** if the time slot is not the last one. <Workload Input Screen>

FIG. **14** is a diagram illustrating an example of a workload input screen **24000**. The workload input screen **24000** is a screen used by the user to newly input a workload. The workload input screen **24000** is provided with a locatable DC base input form **24100** to specify a DC base **1000** at which a workload to be inputted can be located.

The workload input screen **24000** is provided with a delayable time input form **24200** for the user to input a delayable time if the workload to be inputted is a temporal-migration-oriented workload. Furthermore, the workload input screen **24000** is provided with a spatial migratable time input form **24300** to input a spatial migratable time range when the workload is a spatial-migration-oriented workload. The workload input screen **24000** is provided with a spatial non-migratable time input form **24400** to input a spatial non-migratable time range when the workload is a spatial-migration-oriented workload. The workload input screen **24000** is provided with a source code input form **24500** to specify source code of a workload scheduled to be executed at the DC base **1000** and an enter button **24600** to perform workload input.

The user inputs at least one DC base into the locatable DC base input form **24100**. Here, the content inputted is inputted into the executable base **12340** of the workload table **12300** at the DC base **1000** at which the workload is located.

The user inputs a delayable time into the delayable time input form **24200** when the workload to be inputted is a temporal-migration-oriented workload. When the workload execution is not delayable or when the workload is not a temporal-migration-oriented workload, the user inputs 0 or n/a or the user leaves the form blank. The content inputted here is inputted into the delayable time **12335** in the workload table **12300** at the DC base **1000** where the workload is located.

The user inputs a spatial migratable time into the spatial migratable time input form **24300** when the workload to be inputted is a spatial-migration-oriented workload. Although the workload is a spatial-migration-oriented workload, if it cannot be migrated in any time range or when it is not a spatial-migration-oriented workload, the user inputs n/a or leaves the form blank. The content inputted here is inputted into the spatial migratable time range **12345** in the workload table **12300** at the DC base **1000** where this workload is located.

The user inputs a spatial non-migratable time into the spatial non-migratable time input form **24300** when the workload to be inputted is a spatial-migration-oriented workload. Although the workload is a spatial-migration-oriented workload, if it can be migrated in any time range or when it is not a spatial-migration-oriented workload, the user inputs n/a or leaves the form blank. The content inputted here is inputted into the spatial non-migratable time range **12350** in the workload table **12300** at the DC base **1000** where this workload is located.

A path to the source code of the workload scheduled to be executed by the user in the DC base **1000** is specified in the source code input form **24500**. A URL of the source code uploaded to a cloud may be specified or a path of the source code saved in a local storage area may be specified. Instead of specifying a path or a URL, the source code may be directly uploaded.

Once the input is complete, the deployment of the workload starts when the user presses the enter button **24600**.

Effects of Embodiments

In the above-described embodiments, the inter-base workload migratable time range, the workload non-migratable time range and the delayable time of execution time within the same base are managed, and spatial migration whereby a workload is relocated between bases and temporal migration whereby a workload is relocated within the same base are performed in combination. By executing spatial migration of workloads between bases and temporal migration within the same base while taking into account the non-migratable time range of workloads, it is possible to efficiently and more accurately adjust power demand at each base so as to improve a renewable energy utilization ratio in the data center system as a whole. In addition, it is possible to achieve a reduction of power consumption in the data center system as a whole and contribute to environment protection. Adjusting the power demand in the whole data center system can also be used as initiatives for local production and local consumption of renewable energy by efficiently utilizing renewable energy generated in the neighborhood.

In the above-described embodiments, after determining a power amount of power excess to be spatially migrated, a power amount of temporal migration is determined so as to adjust power excess that cannot be adjusted by spatial migration. Thus, it is possible to more accurately adjust power demand at each base.

In the above-described embodiments, it is only when a spatial migratable workload is found in the next time range (time slot) that the power amount of power excess in the target time range is temporally migrated assuming the power migration by spatial migration of the workload as an upper limit. It is possible to prevent the power excess from extending in the time range at the migration destination through temporal migration, thereby resulting in a deterioration of power demand adjustment.

In the above-described embodiments, when an amount of temporally migrating excess power is equal to or smaller than a power adjustment amount due to spatial migration of a workload in the next time range (time slot), part or all of the amount of excess power is temporally migrated. In the above-described embodiments, when the amount of temporally migrating excess power exceeds a power adjustment amount due to spatial migration of a workload in the next time range (time slot), part of the amount of excess power is temporally migrated up to this power adjustment amount. It is thereby possible to prevent the power excess from extending in the time range at the migration destination through temporal migration, thereby resulting in a deterioration of power demand adjustment.

In the above-described embodiments, the spatially migrating power amount is determined, a workload is selected based on the power amount during actual migration and then the workload is spatially migrated. In the above-described embodiments, a power amount to be temporally migrated is determined and then a workload is selected based on the power amount during the actual migration and temporally migrated. Thus, by making a whole migration plan on a power amount basis and selecting individual workloads during the actual migration, it is possible to flexibly and steadily adjust power demand.

The present invention is not limited to the above-described embodiments, and can be implemented using any

given components without departing from the spirit and scope of the invention. The above-described embodiments or modifications are just examples and the present invention is not limited to these contents as long as the features of the invention are not impaired. Although the various embodiments and modifications have been described, the present invention is not limited to these contents. Other aspects conceivable within the scope of technical thought of the present invention are also included within the scope of the present invention.

For example, some of the functions provided in the respective apparatuses of the present embodiments may be provided in other apparatuses or functions provided in other apparatuses may be provided in the same apparatus.

The configurations of the programs described in the present embodiments are just examples, and, for example, part of each program may be incorporated into other programs or a plurality of programs may be configured as one program.

Although a delayable time is used as executable time range, the executable time range may be specified more specifically.

In the present embodiments, although the case with workloads in the data center has been described as workloads, the present invention is also applicable to information processes executed in other facilities or networks.

In the present embodiments, although the purpose of workload control is primarily to improve the renewable energy utilization ratio as described in Description of the Related Art, the present technology can also be utilized as means for providing coordinating power.

Although the present disclosure has been described with reference to example embodiments, those skilled in the art will recognize that various changes and modifications may be made in form and detail without departing from the spirit and scope of the claimed subject matter.

For example, the aforementioned embodiments have been described in detail in order to explain the invention in an easily comprehensible manner and are not necessarily limited to those having all the configurations explained above. Furthermore, part of the configuration of a certain embodiment can be replaced with the configuration of another embodiment and the configuration of another embodiment can be added to the configuration of a certain embodiment. Also, regarding part of the configuration of each embodiment, it is possible to add, delete, replace, integrate, or distribute the configuration. Furthermore, each configuration and each processing illustrated in the embodiments may be distributed or integrated as appropriate based on processing efficiency or implementation efficiency.

What is claimed is:

1. A data center system comprising:

a plurality of bases provided with a calculation resource and a renewable energy power supply to execute a workload; and

an inter-base workload control system to control migration of the workload between the plurality of bases and within a same base, wherein a processor of the inter-base workload control system manages:

a positive excess power amount obtained by subtracting a predicted amount of power supply of the renewable energy power supply from a predicted amount of power consumption of the calculation resource associated with execution of the workload in a future time range at the bases;

spatial migratable time range information on a spatial migratable time range where spatial migration is pos-

sible and the workload scheduled to be executed in the future time range at the bases can be migrated to another base and a non-migratable time range where such migration is not possible;

temporal migratable time range information on temporal migration whereby the workload scheduled to be executed in the future time range at the bases is delayed in execution within the same base to be migrated to another time range; and

a predicted amount of power consumption which is a predicted amount of power consumption of the calculation resource associated with execution of the workload scheduled to be executed in the future time range at the bases,

determines a power adjustment amount indicating the predicted amount of power consumption that moves through the spatial migration and the temporal migration of the workload scheduled to be executed in the future time range at the bases based on the spatial migratable time range information, the temporal migratable time range information and the predicted amount of power consumption so that a sum total of the excess power amounts in the future time range at the plurality of bases becomes smaller, and

executes the spatial migration and the temporal migration of the workload based on the power adjustment amount.

2. The data center system according to claim 1, wherein when the amount of excess power exists in the future time range at the bases, the processor determines the power adjustment amount that moves through the spatial migration of the workload scheduled to be executed in a specific time range at a specific base where the amount of excess power exists so that the sum total becomes smaller, and

after the power adjustment amount is determined, if the amount of excess power further exists in the specific time range at the specific base, the processor determines a first power adjustment amount indicating the predicted amount of power consumption that moves through the temporal migration of the first workload scheduled to be executed in the specific time range at the specific base so that the sum total further becomes smaller.

3. The data center system according to claim 2, wherein the processor determines whether or not there is a second workload scheduled to be executed, the spatial migration of which is possible in the time range next to the specific time range, and

when there is the second workload, the spatial migration of which is possible in the time range next to the specific time range, the processor determines a second power adjustment amount indicating the predicted amount of power consumption that moves through the spatial migration of the second workload as the first power adjustment amount.

4. The data center system according to claim 3, wherein when the amount of excess power in the specific time range is equal to or smaller than the second power adjustment amount, the processor determines the first power adjustment amount so that part or all of the amount of excess power in the specific time range moves to the next time range up to the amount of excess power.

5. The data center system according to claim 3, wherein when the amount of excess power in the specific time range exceeds the second power adjustment amount, the processor

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determines the first power adjustment amount so that part of the amount of excess power in the specific time range moves to the next time range up to the second power adjustment amount.

6. The data center system according to claim 2, wherein the data center system determines the workload to be the spatial migration target in the future time range where the spatial migration is possible based on the power adjustment amount, and executes the spatial migration for the determined workload.
7. The data center system according to claim 2, wherein the data center system determines the workload as the temporal migration target in the future time range where the spatial migration is not possible based on the first power adjustment amount, and executes the temporal migration for the determined workload.
8. An inter-base workload control method executed by an inter-base workload control system, the inter-base workload control system comprising a plurality of bases comprising a calculation resource and a renewable energy power supply to execute a workload to control migration of the workload between the plurality of bases and within a same base, the method comprising steps executed by a processor of the inter-base workload control system managing:
 - a positive excess power amount obtained by subtracting a predicted amount of power supply of the renewable energy power supply from a predicted amount of power consumption of the calculation resource associated with execution of the workload in a future time range at the base,
 - spatial migratable time range information on a spatial migratable time range where spatial migration of migrating the workload scheduled to be executed in the future time range at the base to another base is possible and a non-migratable time range where such spatial migration is not possible,
 - temporal migratable time range information on temporal migration of delaying execution of the workload scheduled to be executed in the future time range at the base within a same base and migrating the workload to another time range and
 - a predicted amount of power consumption which is a predicted amount of power consumption of the calculation resource associated with execution of the workload scheduled to be executed in the future time range at the base,
 - determining a power adjustment amount indicating the predicted amount of power consumption that moves through the spatial migration and the temporal migration of the workload scheduled to be executed in the

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- future time range at the bases based on the spatial migratable time range information, the temporal migratable time range information and the predicted amount of power consumption so that a sum total of the excess power amounts in the future time range at the plurality of bases becomes smaller; and
- executing the spatial migration and the temporal migration of the workload based on the power adjustment amount.
- 9. An inter-base workload control system to control migration of a workload between a plurality of bases and within a same base comprising a calculation resource and a renewable energy power supply to execute the workload, wherein a processor of the inter-base workload control system manages:
 - a positive excess power amount obtained by subtracting a predicted amount of power supply of the renewable energy power supply from a predicted amount of power consumption of the calculation resource associated with execution of the workload in a future time range at the base;
 - spatial migratable time range information on a spatial migratable time range where spatial migration of migrating the workload scheduled to be executed in the future time range at the base to another base is possible and a non-migratable time range where such spatial migration is not possible;
 - temporal migratable time range information on temporal migration of delaying execution of the workload scheduled to be executed in the future time range at the base within a same base and migrating the workload to another time range; and
 - a predicted amount of power consumption which is a predicted amount of power consumption of the calculation resource associated with execution of the workload scheduled to be executed in the future time range at the base,
 - determines a power adjustment amount indicating the predicted amount of power consumption that moves through the spatial migration and the temporal migration of the workload scheduled to be executed in the future time range at the bases based on the spatial migratable time range information, the temporal migratable time range information and the predicted amount of power consumption so that a sum total of the excess power amounts in the future time range at the plurality of bases becomes smaller, and
 - executes the spatial migration and the temporal migration of the workload based on the power adjustment amount.

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